Portfolio Construction and Optimisation using Market Data

Group 4

1 Introduction

Australian superannuation funds manage members' retirement savings around a strategic asset allocation (SAA), the long-run mix of asset classes that the fund targets. APRA's MySuper Heatmap assesses each product's investment performance against benchmark portfolios that reflect its SAA, summarised by its split between growth assets, such as equities and listed property, and defensive assets, such as bonds and cash [1]. Choosing that split quantitatively requires expected returns, volatilities and correlations, which must be estimated from a few decades of noisy data. Because a mean–variance optimiser [10] treats these estimates as exact, errors in them, especially in expected returns, can become large errors in the chosen weights [11]. This tension between optimising and estimating runs through the whole report.

We take the position of an investment team that must build and test an SAA from historical Bloomberg index data on nine asset classes. Our core research question is:

> *How can historical market data be used to construct reliable strategic asset allocations for pension funds, and can tactical reallocation and wealth-dependent multi-period decisions generate additional value?*

Our answer is that historical data support the broad structure of a strategic allocation but not its precise weights. Given the estimates, we can compute optimal portfolios almost exactly; the difficulty lies in the estimates themselves, especially expected returns. Neither monthly tactical reallocation nor wealth-dependent dynamic allocation improved on the fixed strategic allocation in our tests: the first was undermined by noisy return estimates and much higher trading, and the second changed allocations too little over a short horizon to matter. Table 1 summarises how each part of the investigation supports this conclusion.

Table 1: Overview of the investigation in Sections 2 to 5.

Section	Question	Approach	Main finding
2. Inputs	Which data give valid estimates?	Stale-data checks; monthly returns, 2006 to 2026; desmoothing	Hedge-fund NAVs are smoothed (AR(1) coefficient 0.22); desmoothing raises their volatility from 5.3% to 6.6%
3. Strategic allocation	What allocation do the data support?	Frontier; APRA-based SAAs; exponential utility; Monte Carlo; bootstrap	SAA Sharpe ratios 0.50 to 0.52; optimum solved to within 0.75%, but resampling moves it by 42.6%
4. Out of sample and tactical	Do the allocations hold up, and does updating help?	Four 24-month tests; monthly tactical rule, 60–80% growth band	December 2021 portfolios predicted 4.5% to 9.5% a year but returned −0.4% to 1.4%; tactical rule below the SAA in all four tests
5. Dynamic allocation	Does wealth-dependent allocation add value?	Bellman recursion on a wealth grid with Monte Carlo	No measurable gain over a matched static portfolio (CE difference −0.01 basis points)

The study works with asset-class indices rather than a complete model of a fund. It does not model liabilities, contributions, member demographics, fees or taxes, and it treats index returns

as investable. Our conclusions are therefore evidence about how estimation error, constraints and rebalancing rules shape portfolio allocations, not recommendations for a particular fund.

2 Parameter estimation

2.1 Data coverage, effective frequency and returns

The nine Bloomberg indices and their asset classes are listed in Table 2. Their monthly histories begin between January 1997 and December 2005; listed infrastructure starts last and so fixes the start of the common sample. No series has missing observations within its own monthly history.

Higher-frequency data do not always carry new information. HEDGNAV repeats 95.4% of its weekday observations, so it effectively updates monthly. The daily sheet also includes weekends, and markets in different time zones close at different times. As a result, the correlation between Australian and international equities is 0.39 in daily returns but 0.82 in monthly returns, so the sampling frequency changes how much diversification appears to be available.

We therefore use the supplied monthly index levels and simple returns, $r_{i,t} = P_{i,t}/P_{i,t-1} - 1$, without filling missing values. The common sample of index levels begins in December 2005, which gives 248 complete monthly returns from January 2006 to August 2026.

2.2 Annualised parameters and covariance validity

Hedge-fund returns exhibit first-order autocorrelation $\hat{\rho} = 0.219$, consistent with smoothed NAV reporting [7]. For estimation, we apply the desmoothing method of Geltner [6], $r^*_{H,t} = (r_{H,t} - \hat{\rho} r_{H,t-1})/(1 - \hat{\rho})$, retaining the observed first return. This raises annualised hedge-fund volatility from 5.28% to 6.58%. Other series remain unadjusted; original returns are retained for realised performance evaluation.

Let r^*_t denote the adjusted nine-asset return vector. With $T = 248$, the annualised estimates are $m = 12\bar{r}^*$ and $C = 12\widehat{\mathrm{Cov}}(r^*_t)$, using the sample covariance with denominator $T-1$. Scaling by 12 assumes that monthly autocovariances are negligible. Cash is kept as an asset with its observed variance, and its annualised mean of 3.289% is also used as the risk-free rate in Sharpe ratios.

Table 2: Annualised parameter estimates, January 2006–August 2026. Returns and volatilities are percentages.

Index	Asset class	Mean	Volatility
ASA52	Australian equities	8.183	13.800
NDDLWI	International equities	9.571	14.095
NDLEEGF	Emerging equities	9.602	16.096
RDAU	Australian property	4.744	19.963
FDCIISAH	Listed infrastructure	9.516	11.458
HEDGNAV	Hedge funds (adjusted)	5.431	6.580
BACM0	Australian bonds	4.035	3.737
H03432AU	Global bonds	4.261	3.472
BAUBIL	Cash	3.289	0.565

The estimated covariance matrix is exactly symmetric and positive definite, with smallest eigenvalue 2.86×10^{-5}, so no correction is needed. Its condition number is about 3,063, however, because cash has very low variance and the bond exposures are correlated. Unconstrained weights can therefore react strongly to small errors in the inputs.

2.3 Stability and implications for portfolio construction

To check stability, we compare estimates from the full sample, 2006–2015, the final 120 months and the final 60 months, together with 60-month rolling estimates. These descriptive comparisons use the full-sample desmoothing coefficient; the backtests in Section 4 re-estimate it in each window. Expected returns vary widely. The Australian bond mean falls from 6.063% in 2006–2015 to 0.015% in the final five years, while the international equity mean rises from 5.896% in 2006–2015 to 13.468% in the final ten years.

Correlations and volatilities change too. The correlation between international equities and global bonds rises from -0.270 in 2006–2015 to 0.655 in the final five years, so bonds offered much less diversification against equity risk in the recent period. Australian bond volatility rises from 2.831% to 5.387% across the same periods.

If monthly returns are iid, the standard error of m_i is $\sqrt{12C_{ii}/T}$. For the three equity indices this is 3.04 to 3.54 percentage points, which is large next to estimated means of 8.2% to 9.6%. Serial dependence, changing volatility and structural breaks make this only an approximation, and scaling the covariance by 12 is also approximate for persistent series such as cash.

Uncertain means and changing correlations motivate the allocation constraints, sensitivity analysis, bootstrap and out-of-sample tests that follow.

3 Strategic portfolio construction and reliability

3.1 Constraints and the efficient frontier

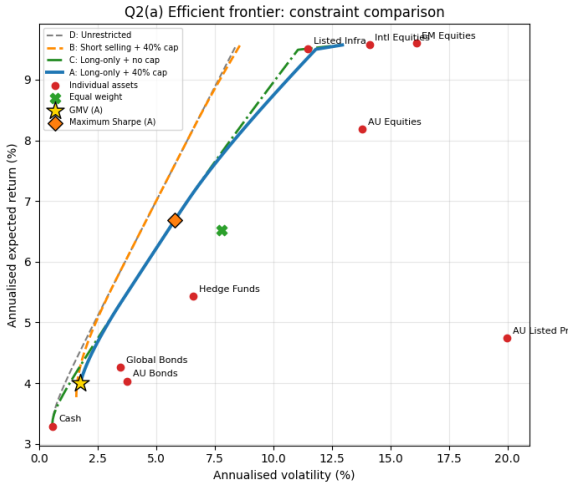
Short selling is permitted, so we compare four fully invested sets ($\mathbf{1}^\top x = 1$): A, $0 \le x_i \le 0.40$; B, $x_i \le 0.40$ with short selling and no explicit lower bound; C, long-only without the cap; and D, unrestricted weights. For each set $\mathcal{F}$, using the annualised estimates of Section 2,

$$\min_{x \in \mathcal{F}} x^\top C x \qquad \text{subject to} \qquad x^\top m \ge \mu.$$

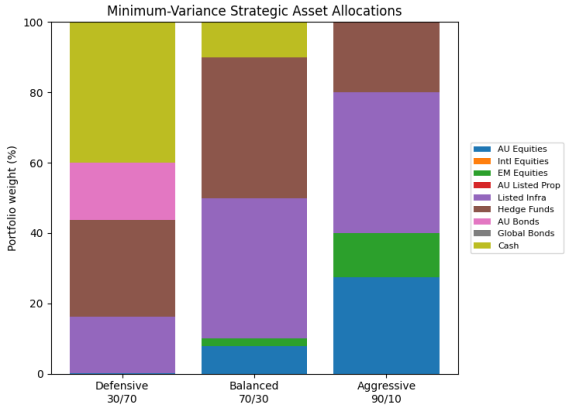
Figure 1a isolates the short-selling restriction through A versus B, and the cap through A versus C. Short selling reduces estimated risk over much of the plotted range, but these gains exclude borrowing costs. We retain A for the rest of the report to avoid leveraged long–short positions and limit concentration. The 40% cap requires at least three holdings; it is a modelling choice, not a requirement of the project. Both restrictions also limit extreme positions driven by uncertain means and an ill-conditioned covariance matrix (condition number 3,063; Section 2.2).

Under A, the global minimum-variance (GMV) portfolio holds 40.00% cash, 24.13% global bonds, 21.14% Australian bonds and 14.73% hedge funds, with an expected return of 4.00% and volatility of 1.76%. Without the cap, the long-only GMV portfolio holds approximately 99% cash with 0.56% volatility. Targets of 6% and 8% require volatilities of 4.62% and 8.32%; higher targets shift holdings towards infrastructure and equities. The maximum Sharpe ratio is 0.587 at return 6.69% and volatility 5.80%, using the mean cash return of 3.29% as the risk-free rate.

Targets below the GMV return leave the return constraint slack; higher targets bind up to A's maximum attainable return of 9.57%. Feasibility, KKT and CVXPY [4] checks pass, and relaxing the bounds recovers the analytical solution. Yet re-estimating with only the last ten or five years moves the 6%-target portfolio by 72.8% and 70.7%, versus 14.0% and 14.8% for the GMV portfolio, measured by the half-L^1 distance $D(x, y) = \frac{1}{2}\sum_i |x_i - y_i|$, the fraction of the portfolio that must be traded to move from one to the other. Mean-only changes have larger effects than covariance-only changes in these window comparisons, so numerical accuracy does not ensure allocation stability.



(a) Four constraint specifications.



(b) Strategic portfolio weights.

Figure 1: Full-sample portfolio construction with hedge-fund desmoothing. All portfolios are fully invested; panel (b) uses specification A and growth targets of 30%, 70% and 90%.

3.2 Strategic allocations and diversification

Following APRA's classification [1, Table 4, p. 9], equities, listed property and infrastructure are growth assets, while bonds and cash are defensive. We assign hedge funds to APRA's 'Other' category, counting them as 50% growth and 50% defensive. With growth shares g and $\mathcal{F}_A$ the feasible set of specification A, the defensive, balanced and aggressive SAAs solve

$$\min_{x \in \mathcal{F}_A} x^\top C x \quad \text{subject to} \quad g^\top x = q, \qquad q \in \{0.30, 0.70, 0.90\}.$$

Their weights therefore depend on the covariance matrix and the growth exposure, not on estimated means.

Figure 1b shows the binding caps: the defensive portfolio holds 40% cash; the balanced portfolio holds 40% each in infrastructure and hedge funds, with 10% each in cash and equities; and the aggressive portfolio holds 40% infrastructure, 20% hedge funds and 40% equities. In the same order, expected returns are 5.01%, 7.15% and 8.34%, volatilities 3.32%, 7.58% and 10.07%, and Sharpe ratios 0.519, 0.510 and 0.502.

The diversification ratio, $\sum_i x_i \sqrt{C_{ii}} / \sqrt{x^\top C x}$, equals 1.36, 1.15 and 1.16: correlations below one reduce portfolio volatility relative to the weighted-average asset volatility. However, fixing the growth exposure costs approximately 0.34, 0.97 and 1.00 percentage points of volatility relative to frontier portfolios from Section 3.1 with the same returns and asset bounds.

Classification materially affects the result. Counting hedge funds as 100% growth changes the aggressive portfolio's expected return and volatility to 7.15% and 7.58%, essentially reproducing the original balanced allocation. Growth labels therefore do not uniquely determine risk. The defensive portfolio prioritises lower volatility and nearer-term cash needs, the aggressive portfolio requires greater loss tolerance and a longer horizon, and the balanced portfolio is intermediate. Their similar Sharpe ratios provide no decisive ranking, which motivates the preference-based analysis that follows.

3.3 Risk preferences and numerical optimisation

The strategic allocations above fix the growth exposure; we now let the investor's risk preferences choose the allocation. For an investor with initial wealth $W_0 = 1$, exponential utility $U(W) = -e^{-\gamma W}$ with absolute risk aversion $\gamma > 0$ and normal annual returns $R \sim N(m, C)$, terminal wealth $W_1 = 1 + x^\top R$ is normal, and the moment generating function gives $\mathbb{E}[U(W_1)] =$

$-\exp\{-\gamma(1 + x^\top m - \frac{\gamma}{2} x^\top C x)\}$. Maximising expected utility is therefore equivalent to

$$\max_x \; x^\top m - \frac{\gamma}{2} x^\top C x, \tag{1}$$

the certainty-equivalent (CE) return, subject to the constraints of specification A (Section 3.1). Since (1) is a positive multiple of $t\, x^\top m - \frac{1}{2} x^\top C x$ with risk tolerance $t = 1/\gamma$, every optimum lies on the efficient frontier; numerically, the multiplier of the frontier's return constraint equals $2/\gamma$ to a relative error below 10^{-6}.

Effect of risk aversion. As γ rises from 1 to 100 (Figure 2(a)), the expected return falls from 9.56% to 4.03% and volatility from 11.95% to 1.77%. For $\gamma \leq 3$ the portfolio sits at the maximum-return corner (40% infrastructure, the rest in international and EM equities); defensive assets then enter one at a time, and as $\gamma \to \infty$ the optimum converges to the GMV portfolio. Australian equities and listed property are never held. The growth exposure is 75.9%, 61.1%, 47.6% and 32.0% at $\gamma = 5$, 7, 10 and 15, so the balanced allocation of Section 3.2 corresponds to γ between 5 and 7 and the defensive one to $\gamma \approx 15$. The Sharpe ratio peaks at about 0.59 for γ between 10 and 15, and the $\gamma = 10$ portfolio (expected return 6.72%, volatility 5.85%) is close to the maximum-Sharpe portfolio, so we use $\gamma = 10$ as the reference investor.

Monte Carlo optimisation. Because (1) relies on normality, we also solve the problem by simulation: we draw S scenarios $R_s \sim N(m, C)$ once and maximise the sample average of $-e^{-\gamma(1+x^\top R_s)}$ in log-sum-exp form, reusing the same draws for every candidate x so that the objective is smooth. This isolates the error in *solving* the model from the error in *estimating* m and C (Section 3.4). Over 20 runs per sample size at $\gamma = 10$ (Figure 2(b)), the distance D to the exact optimum falls from 30.3% at $S = 250$ to 1.5% at $S = 256{,}000$, and from 18.6% to 0.75% with antithetic variates [8], which pair each draw z with $-z$ and cut the variance of the utility estimate by a factor of 0.28. As theory predicts, D decays like $S^{-1/2}$ (slopes -0.44 and -0.45) while the CE loss decays like S^{-1} (slopes -0.97 and -0.94), because expected utility is flat near its maximum; at $S = 256{,}000$ the loss is only 0.006 basis points. The closed form is far faster under normality (a few hundredths of a second against about 0.7 seconds for the antithetic run at $S = 256{,}000$), but simulation needs neither normal returns nor a particular utility function, which is why Section 5 uses it.

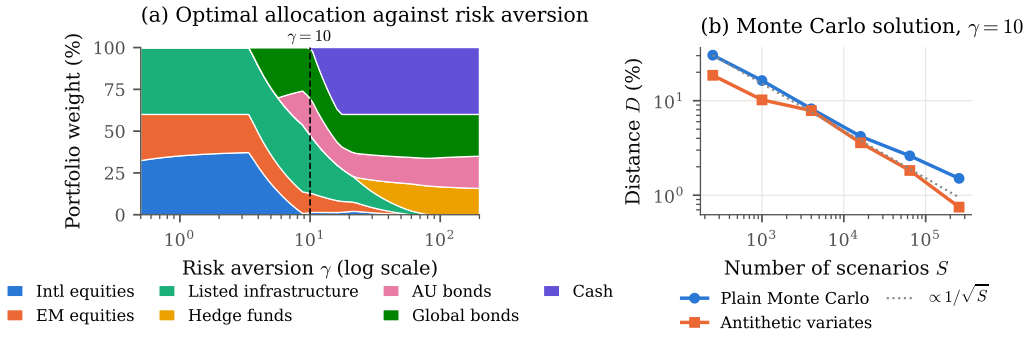


Figure 2: (a) Optimal weights of the constrained CARA–normal problem against risk aversion γ (log scale). (b) Mean distance D between the Monte Carlo solution and the exact optimum at $\gamma = 10$, over 20 runs per sample size, with a $1/\sqrt{S}$ reference line.

3.4 Sampling uncertainty and allocation reliability

To measure how much the optimum depends on the particular 248 months observed, we resample whole months with replacement [5], re-estimate m and C, and re-solve (1) at $\gamma = 10$, 1,000

times. This keeps the dependence between assets but not serial dependence such as volatility clustering.

Table 3: Optimal weights at $\gamma = 10$: full-sample point estimate and distribution over 1,000 bootstrap samples. Australian and international equities and listed property are not shown.

Asset	Point estimate	Bootstrap mean	90% interval	Held in
Listed infrastructure	34.7%	23.8%	0–40%	84%
Global bonds	30.3%	19.8%	0–40%	71%
Australian bonds	22.1%	18.3%	0–40%	68%
EM equities	11.6%	10.1%	0–35.0%	60%
Cash	0.0%	10.8%	0–40%	41%
Hedge funds	0.0%	5.7%	0–39.6%	26%

The individual weights are poorly determined (Table 3). Each 90% interval runs from zero to near the 40% cap, cash and hedge funds are held in 41% and 26% of samples despite being absent from the point estimate, and a bootstrap optimum lies on average at $D = 42.6\%$ from the point estimate. Most of this comes from the expected returns: resampling only m moves the portfolio by 40.5% on average, against 17.7% for resampling only C (the distances are not additive), consistent with the unstable means of Section 2 and with the finding that errors in means matter far more than errors in covariances [3]. Optimisation also exploits sampling noise: the bootstrap optima promise 7.71% on their own samples but only 6.52% under the full-sample estimates. The instability is smaller at the extremes, $D = 35.3\%$ at $\gamma = 3$ (a corner solution) and 26.3% at $\gamma = 30$ (near the GMV portfolio, which does not depend on m).

Given m and C, then, the model is solved almost exactly ($D = 0.75\%$), yet equally plausible samples move the optimum by $D = 42.6\%$: reliability is limited by the data, not the numerical method. Individual weights should not be taken literally; only the broad structure, a mix of growth and defensive assets with no listed property, is robust. Section 4 therefore tests these allocations on returns not used to construct them.

4 Out-of-sample and tactical portfolio allocation

4.1 Historical design and strategic benchmarks

We conduct four separate historical experiments, formed in December 2015, 2018, 2021 and 2023. Each uses the 120 monthly observations ending at the formation date and a subsequent 24-month test: 2016–2017, 2019–2020, 2022–2023 and 2024–2025, respectively. The ten-year window balances sample size against changing market conditions, and the tests are not joined into a continuous strategy.

At each formation date T_0, the annualised means and covariances, m_{T_0} and C_{T_0}, are re-estimated using the procedure of Section 2, with hedge-fund desmoothing re-estimated within each window. We assume that observations, including hedge-fund NAV returns, are available at their recorded month-end and can inform the following month's allocation. This avoids look-ahead bias, which would flatter the backtest, although actual reporting delays or revisions could affect implementability.

The defensive, balanced and aggressive portfolios are reconstructed using the minimum-variance problem, growth targets and investment constraints of Section 3.2. Their target weights remain fixed and are restored monthly, rather than allowed to drift as in buy-and-hold. Realised performance uses the original reported returns, and desmoothing is used only for estimation. Baseline results exclude transaction costs.

4.2 Predicted and realised strategic performance

For fixed target weights x, realised monthly returns are $r_{p,t} = x^\top r_t$, using original reported returns. Predicted and realised annual statistics are

$$\widehat{\mu}_p = x^\top m_{T_0}, \qquad \widehat{\sigma}_p = \sqrt{x^\top C_{T_0} x}, \qquad \widehat{S}_p = \frac{\widehat{\mu}_p - \widehat{r}_f}{\widehat{\sigma}_p},$$

$$\mu_p^{\text{real}} = 12\bar{r}_p, \qquad \sigma_p^{\text{real}} = \sqrt{12}\, s(r_p), \quad S_p^{\text{real}} = \frac{12(\bar{r}_p - \bar{r}_{\text{cash}})}{\sigma_p^{\text{real}}}.$$

Here s is the sample standard deviation. Predicted Sharpe ratios use the training-period annualised mean cash return $\widehat{r}_f$, and realised Sharpe ratios use evaluation-period cash returns. Annual returns are arithmetic means, not compound growth rates. For $H = 24$ months, the cumulative return is $R_{\text{cum}} = \prod_{t=1}^{H}(1 + r_{p,t}) - 1$.

Figure 3a shows that the forecasts miss in both directions. After December 2018, all three portfolios were more volatile than predicted, the aggressive portfolio reaching 15.48% against 7.94%. The largest return shortfalls came after December 2021: the defensive, balanced and aggressive portfolios were predicted to return 4.48%, 7.82% and 9.46% a year but realised 1.42%, −0.03% and −0.38%, with cumulative returns of 2.76%, −0.81% and −2.17%, all below cash. The December 2023 portfolios did the opposite, with higher returns and lower volatility than predicted; the balanced portfolio returned 11.36% a year and 25.16% in total.

In every test, more growth exposure meant higher realised volatility but not consistently higher returns. Predicted risk uses desmoothed hedge-fund returns while realised risk uses reported returns, so the gaps reflect measurement as well as market conditions, and Sharpe comparisons are also affected by changes in cash returns.

Figure 3b shows how the allocations change between formation dates with the growth targets held fixed. Between December 2015 and December 2018, the distance D is 29.57%, 39.21% and 46.83% for the defensive, balanced and aggressive portfolios, partly because hedge-fund weights increase. The constraints are unchanged and these are minimum-variance portfolios, so the changes come from the covariance estimates. In contrast, the December 2021 and December 2023 aggressive portfolios differ by only $D = 0.51\%$, yet their cumulative returns were −2.17% and 30.29%. A stable allocation does not mean stable performance. With only four tests the evidence is limited, which raises the question of whether updating the estimates every month helps.

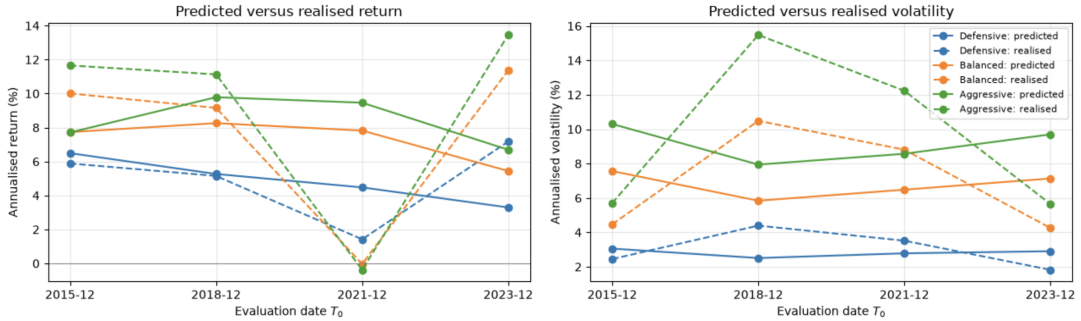
4.3 Tactical strategy and implementation

The tactical asset allocation (TAA) keeps the balanced SAA as its reference and lets growth exposure move between 60% and 80%, so it can tilt around the 70% target without leaving the balanced mandate. At each month-end k it solves

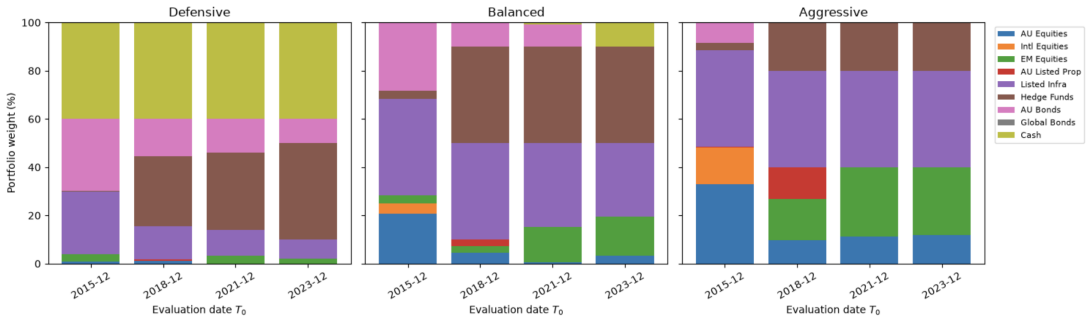
$$\max_x \; x^\top m_k - \frac{\gamma}{2} x^\top C_k x \quad \text{subject to} \quad \mathbf{1}^\top x = 1, \quad 0 \le x_i \le 0.40, \quad 0.60 \le g^\top x \le 0.80,$$

which is problem (1) with a growth band added. We use $\gamma = 10$, the reference investor of Section 3.3, and re-estimate m_k and C_k each month from the preceding 120 observations, so the estimation window and rebalancing frequency match the strategic benchmark. Using monthly instead of annual parameters would divide the objective by 12 and leave the solution unchanged.

The evaluation periods, window-specific desmoothing and information assumption are those of Section 4.1, and weights chosen at a month-end earn only the following month's reported returns. The solutions pass feasibility and optimality checks, and our implementation reproduces the fixed balanced benchmark of Section 4.2. TAA is a rolling one-period rule with wealth normalised to one, not a wealth-dependent policy.



(a) Predicted and realised annual arithmetic return and volatility.



(b) SAA weights at the four formation dates.

Figure 3: Strategic allocations in four separate 24-month tests, with growth targets of 30%, 70% and 90% (hedge funds count as half growth). Estimation uses 120 months and window-specific hedge-fund desmoothing; evaluation uses reported returns, fixed targets are restored monthly and returns exclude costs. Weight differences in panel (b) are target changes, not realised turnover.

4.4 Performance, weight changes and sensitivity

TAA ends each baseline test with less wealth than the fixed SAA (Table 4). Its volatility is higher in three of the four periods and its maximum drawdown is larger in all four; in 2019–2020 the drawdown reaches 20.21%, against 14.32% for the SAA. Its only Sharpe-ratio improvement is marginal, 1.819 against 1.815 in 2016–2017.

Table 4: Baseline results, all in per cent. Returns are cumulative over 24 months and gross of costs; volatility is annualised. Turnover is total one-way trading over each test, excluding initial establishment.

	Cumulative return		Annual volatility		Total turnover	
Test period	TAA	SAA	TAA	SAA	TAA	SAA
2016–2017	21.29	21.83	4.34	4.47	80.15	13.10
2019–2020	10.36	18.72	14.01	10.48	192.47	16.08
2022–2023	−7.45	−0.81	11.54	8.81	346.75	21.46
2024–2025	24.83	25.16	4.66	4.26	169.43	14.56

The weight changes follow movements in the estimated means and risks. In October 2017, the international equity weight rose from 12.63% to 20.00% while the EM equity weight fell to zero. Both estimated means declined, but the EM mean fell further, from 6.09% to 5.30%, against 5.80% to 5.73% for international equities. In April 2020, the listed property weight fell from 25.21% to zero as its estimated mean dropped from 11.78% to 7.89%, its volatility rose from 11.77% to 17.24% and its correlation with international equities rose from 0.322 to 0.465. These comparisons help explain the changes but do not isolate the effect of each input.

Shortening the estimation window to 60 months moves the average allocation by $D = 16.13\%$ to 51.81% across the tests. It raises TAA's cumulative return in all four tests but beats the

SAA in only two, by 0.46 and 2.92 percentage points, and it does not reduce risk uniformly: volatility in 2016–2017 rises from 4.34% to 6.74%.

With the 120-month window, TAA with $\gamma = 5$ beats the SAA in two tests and with $\gamma = 20$ in none. In 2019–2020, raising γ from 5 to 20 cuts realised volatility from 17.81% to 10.96%, but both versions underperform the SAA. These checks show how sensitive the strategy is to its settings; choosing a setting after seeing its performance would not be a fair test.

4.5 Turnover, costs and tactical value

Monthly turnover is $\tau_k = D(x_k, x_k^-)$, where x_k^- is the portfolio after the previous month's returns have moved the weights. The fixed SAA also trades, since drifting weights must be restored to target. Over each two-year test, baseline TAA turnover is 80.15% to 346.75%, against 13.10% to 21.46% for the SAA. For an illustrative cost c per dollar bought or sold, each strategy pays for its own turnover:

$$r_k^{\text{net}} = (1 - 2c\tau_k)(1 + r_k^{\text{gross}}) - 1.$$

Costs are deducted in proportion before each month's return, targets are unchanged, and the initial purchase of the portfolio is not charged. At 10 basis points, TAA's cumulative-return shortfall against the SAA widens to -0.70, -8.75, -7.24 and -0.72 percentage points. At 25 basis points, the 60-month strategy's small 2016–2017 gain over the SAA becomes a 0.16-point loss, although its 2024–2025 gain stays positive.

These tests show no consistent tactical advantage: noisy estimates of the means and the extra trading both work against monthly reallocation. The comparison also changes the objective and the growth constraint, though, so it does not isolate the effect of updating the parameters. Four historical tests and illustrative costs are thin evidence, and any refinement, such as more stable estimates or a turnover penalty, would need its own out-of-sample test.

5 Multiperiod dynamic portfolio optimisation

5.1 Fixed, tactical and dynamic decisions

The fixed SAA rebalances to constant weights, and TAA updates its market estimates and solves a one-period utility problem within its tactical limits. The dynamic strategy instead lets the allocation depend on current wealth and the time remaining, and maximises the expected utility of terminal wealth. In each dynamic experiment, monthly returns are iid with parameters estimated once at the start and then held fixed, so any reallocation is driven by wealth and time rather than by new market estimates.

5.2 Bellman formulation and wealth dependence

Starting from $W_0 = 1$, the weights x_k are chosen before R_{k+1} is observed, and wealth evolves as

$$W_{k+1} = W_k(1 + x_k^\top R_{k+1}), \qquad \mathcal{X} = \{x \in \mathbb{R}^9 : \mathbf{1}^\top x = 1,\ 0 \le x_i \le 0.4\}. \tag{2}$$

The basic model has no growth-exposure constraint; a version restricted to TAA's growth band is included in the comparison of Section 5.4.

For exponential utility $U(w) = -e^{-\gamma w}$, let $V_k(w)$ be the maximum expected terminal utility attainable from wealth w at date k. Because returns are independent, V_k satisfies the Bellman recursion [2]

$$V_N(w) = -e^{-\gamma w}, \qquad V_k(w) = \max_{x \in \mathcal{X}} \mathbb{E}\big[V_{k+1}\big(w(1 + x^\top R_{k+1})\big)\big]. \tag{3}$$

The resulting policy $x_k = \pi_k(W_k)$ takes into account that the portfolio can be changed again at later dates.

At the final decision date, suppose $w > 0$ and $R_N \sim N(m_\Delta, C_\Delta)$, where m_Δ and C_Δ are the monthly parameters (the annualised estimates divided by 12). Terminal wealth then has mean $w(1 + x^\top m_\Delta)$ and variance $w^2 x^\top C_\Delta x$, and the normal exponential-moment formula gives

$$\mathbb{E}[U(W_N) \mid W_{N-1} = w] = -\exp\left[-\gamma w(1 + x^\top m_\Delta) + \frac{\gamma^2 w^2}{2} x^\top C_\Delta x\right]. \tag{4}$$

Maximising this is equivalent to maximising the certainty-equivalent wealth $w(1 + x^\top m_\Delta) - \gamma w^2 x^\top C_\Delta x/2$. Dividing by $w > 0$ and removing the constant gives

$$x^*_{N-1}(w) = \arg\max_{x \in \mathcal{X}} \left\{ x^\top m_\Delta - \frac{\gamma w}{2} x^\top C_\Delta x \right\}. \tag{5}$$

This is problem (1) in monthly units with effective risk aversion γw. Absolute risk aversion stays at γ, but relative risk aversion γw rises with wealth. With fixed parameters and constraints, the optimal portfolio's variance and expected return are therefore non-increasing in wealth, although individual weights need not move monotonically.

As an illustration we use the 120 months to December 2015, with window-specific desmoothing and $\gamma = 10$. At wealth 0.50, 0.75 and 1.00, the final-period allocation is the same: 40% infrastructure, 20% Australian bonds and 40% global bonds. At wealth 2.00, infrastructure falls to 23.95% and Australian bonds rise to 36.05%, and monthly volatility falls from 1.259% to 0.890%. Lower percentage risk does not mean lower risk in dollars, however: the standard deviation of terminal wealth rises from 0.00630 at wealth 0.50 to 0.01780 at wealth 2.00.

These are final-period model results, not out-of-sample performance. Earlier decisions depend on the continuation value in (3), so applying (5) at every date does not in general solve the multiperiod problem.

5.3 Numerical solution and validation

We solve the Bellman problem by backward induction over $N = 3$ monthly investment periods, using the December 2015 calibration of Section 5.2, $\gamma = 10$ and $W_0 = 1$. Monthly returns are independent normal vectors with fixed parameters (m_Δ, C_Δ). The short horizon keeps the problem tractable and allows direct validation against the final-period solution. Wealth is discretised on 51 equally spaced points in $[0.5, 1.5]$, a spacing of 0.02. To avoid interpolating utilities close to zero, we store certainty-equivalent terminal wealth $J_k(w) = -\gamma^{-1} \log[-V_k(w)]$, with $J_N(w) = w$, and approximate the Bellman step at each date and grid point by

$$\widehat{J}_k(w) = \max_{x \in \mathcal{X}} \left\{ -\frac{1}{\gamma} \log \left[\frac{1}{M} \sum_{s=1}^{M} \exp\left\{ -\gamma \widetilde{J}_{k+1}\left(w(1 + x^\top R^{(s)}_{k+1}) \right) \right\} \right] \right\},$$

where $M = 20{,}000$ and $\widetilde{J}_{k+1}$ linearly interpolates the stored continuation values, with linear extrapolation beyond the grid. Each date uses its own random stream, held fixed across candidate portfolios and wealth states, so the optimiser sees no resimulation noise. The logarithm is evaluated in log-sum-exp form and each problem is solved with SLSQP [9], warm-started from the neighbouring wealth state; every solution satisfies full investment and the long-only 40% limits. Forward evaluation interpolates the weights and uses the nearest boundary policy outside the grid.

The baseline initial certainty equivalent is $\widehat{J}_0(1) = 1.017964$. The initial allocation is 40% listed infrastructure, 20% Australian bonds and 40% global bonds, with expected monthly return 0.673% and volatility 1.259%. At wealth one the allocation barely changes across decision dates: the infrastructure weight is 40.00%, 39.34% and 40.00% at $k = 0, 1, 2$, while global bonds stay at their cap.

The numerical final-period policy matches the analytical benchmark at grid wealth levels 0.74 and 1.00, and the largest individual-weight error, at 1.24, is 0.45 percentage points. With $M = 20{,}000$, refining the grid from 31 to 51 and 71 points changes the initial certainty equivalent by less than 2×10^{-7} and leaves the initial weights unchanged. With 51 grid points, raising the simulation size from 5,000 to 40,000 gives initial certainty equivalents between 1.017731 and 1.018025; for 10,000 or more draws the initial weights coincide and the CE differs from the 40,000-draw value by less than 4×10^{-5}. These checks show limited sensitivity near the initial state, but they do not bound the approximation error over the whole policy.

On 20,000 independent forward paths, the estimated certainty equivalent is 1.018082. Its difference from backward induction, 0.000118, is smaller than its Monte Carlo standard error of 0.000158, and no simulated wealth state leaves the grid. This supports the numerical solution and the grid's coverage for this calibration, although it does not prove global optimality or rule out rare boundary events.

5.4 Historical and model-based comparisons

We compare the policies over January–March 2016, 2019, 2022 and 2024, following the four formation dates of Section 4.1. Each dynamic model uses the preceding 120 monthly observations, with window-specific hedge-fund desmoothing, and holds its estimates fixed over the three-month horizon. Realised performance uses the original reported returns. The fixed balanced SAA restores its target weights monthly, TAA uses the rolling allocations of Section 4.3, and the dynamic policies respond to current wealth and the remaining horizon. Every decision precedes the returns it earns, under the month-end information assumption of Section 4.1.

The feasible sets differ: the fixed SAA requires 70% growth exposure, TAA permits 60–80%, and the baseline dynamic policy has no growth restriction. We therefore add a second dynamic policy with TAA's 60–80% growth limits. All four strategies are fully invested, long-only and subject to the 40% asset cap. Matching the growth limits improves comparability, although TAA still updates its parameters and uses a one-period objective.

Table 5: Cumulative returns (%) over identical three-month historical periods, before transaction costs. Each row is a separate experiment.

Test period	Fixed SAA	TAA	Dynamic	Dynamic (60–80%)
Jan–Mar 2016	2.59	4.41	4.73	4.43
Jan–Mar 2019	8.78	10.14	8.98	9.03
Jan–Mar 2022	0.75	−1.73	−1.89	−1.89
Jan–Mar 2024	3.84	5.44	4.14	5.19

No strategy delivers the highest return in every experiment (Table 5). The baseline dynamic policy leads in early 2016, TAA in 2019 and 2024, and the fixed SAA in 2022. The 2016 dynamic portfolio starts with only 40% growth exposure, against 60% for TAA and 70% for the SAA. Imposing TAA's growth range lowers its return from 4.73% to 4.43% and raises its monthly volatility from 0.67% to 1.96%, close to TAA's 1.99%, so the baseline's lower realised risk partly reflects its different mandate. In 2022 the growth limits leave the two dynamic policies effectively unchanged. Binding asset caps and small wealth changes also limit how much a wealth-dependent policy can adjust over three months.

To measure expected terminal utility separately, we simulate 50,000 independent three-month paths under the December 2015 normal model, using common paths across strategies. Each dynamic policy is compared with a static mean–variance portfolio computed at $W_0 = 1$ under identical parameters and constraints and rebalanced monthly. These controls isolate the combined value of wealth- and horizon-dependent decisions relative to a fixed utility-based allocation, which is not the same as rolling TAA.

Simulated certainty-equivalent terminal wealth is 1.017895 for the baseline dynamic policy and 1.012235 for the fixed SAA, but the matched static control achieves 1.017896. The paired dynamic-minus-static CE difference is -0.0097 basis points of initial wealth, with an approximate 95% Monte Carlo interval of $[-0.0375, 0.0181]$. With the growth band, the dynamic and static CEs are 1.015451 and 1.015454, a difference of -0.0267 basis points with interval $[-0.0806, 0.0272]$. Both intervals contain zero. They measure simulation uncertainty conditional on the fitted model and exclude parameter uncertainty and dynamic-programming approximation error, so the small negative estimates do not show that the exact optimal dynamic policy is inferior.

Trading can also offset any benefit. In early 2024, drift-adjusted one-way turnover, excluding initial establishment, is 35.43% for the baseline dynamic policy, 23.50% for TAA and 1.46% for the fixed SAA, and the historical returns above exclude these costs. With only three realised monthly returns per experiment, the volatility estimates are descriptive. The historical tests show no consistent performance advantage, and the matched simulations show no clear utility gain from dynamic reallocation at this short horizon and calibration.

6 Discussion and conclusion

Historical data can guide a pension fund's strategic allocation, but only at the level of its broad structure. Solving the optimisation is not the obstacle, since Monte Carlo weights sit within $D = 0.75\%$ of the closed form; estimation is. Equally plausible samples move the $\gamma = 10$ optimum by $D = 42.6\%$, mostly through the means, and the optimised portfolios promise 7.71% in sample against 6.52% under the full-sample estimates. Out of sample, the December 2021 portfolios were predicted to return 4.5% to 9.5% a year and returned -0.4% to 1.4%. What survives is the shape of the allocation, a growth–defensive mix with infrastructure as the main growth holding and no listed property. The near-equal Sharpe ratios of the three SAAs (0.50 to 0.52) show that the growth target is a choice about risk tolerance, not something the data can optimise. Extra flexibility did not add value in our tests. Monthly tactical reallocation trailed the fixed SAA in all four periods while trading 6 to 16 times as much, and the wealth-dependent policy's certainty equivalent differed from a matched static portfolio by -0.01 basis points, an amount indistinguishable from zero.

For a fund, the practical lesson is to set the strategic mix through the growth–defensive split and members' risk tolerance, to read optimised weights as ranges rather than targets, and to use constraints such as the 40% cap as protection against estimation error. Tactical or dynamic rules should be adopted only if they beat a fixed allocation after costs out of sample, which none of ours did. These conclusions have limits. The evidence rests on about twenty years of data, four non-overlapping tests and illustrative costs, and mean estimates shift between regimes. Desmoothing is approximate, and the whole-month bootstrap and normal-return models ignore volatility clustering and fat tails. The backtests assume month-end data are available immediately, and the dynamic problem covers only three months with fixed parameters. The tactical and dynamic comparisons also change the objective or growth constraint, not only the updating rule. Our results therefore do not show that fixed allocations always dominate or that dynamic optimisation is ineffective in general. The most useful next steps are to reduce estimation error directly, for example by shrinking the mean estimates or penalising turnover, and to test longer-horizon dynamic policies with time-varying parameters, using a block bootstrap that preserves serial dependence.

Acknowledgement of AI use

Claude was used to optimise and latexify the content solved in the report and to remove bugs from the code.

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